UNIVERSITI TEKNOLOGI MARA
MARKING SCHEME

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|---|---|---|
| COURSE | : | COMPUTER ORGANIZATION AND ARCHITECTURE |
| COURSE CODE | : | CSC429 |
| EXAMINATION | : | NOVEMBER 2026 (MOCK) |
| TIME | : | 3 HOURS |

MARKING SCHEME NOTES

1. This marking scheme accompanies the CSC429 mock paper (November 2026). All answers are computed deterministically. Worked answers are shown for data representation and assembly; descriptive marking points come from a fixed bank.

2. No placeholders are used; every slot has a worked or descriptive answer.

3. Part A = 10 questions (80 marks); Part B = 1 assembly question (20 marks); total = 100 marks.

4. Computed slots: 4; descriptive slots: 7; placeholders: 0. Questions and answers are for local study use only.

5. Fingerprint based on 20 unique CSC429 papers; output is deterministic for seed 15.

FOR MARKERS' USE ONLY

*This marking scheme consists of 9 printed pages.*

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## PART A — MARKING SCHEME

Worked answers and marking points.

### QUESTION A1 (5 marks)

Study Figure 1 and answer the following questions.

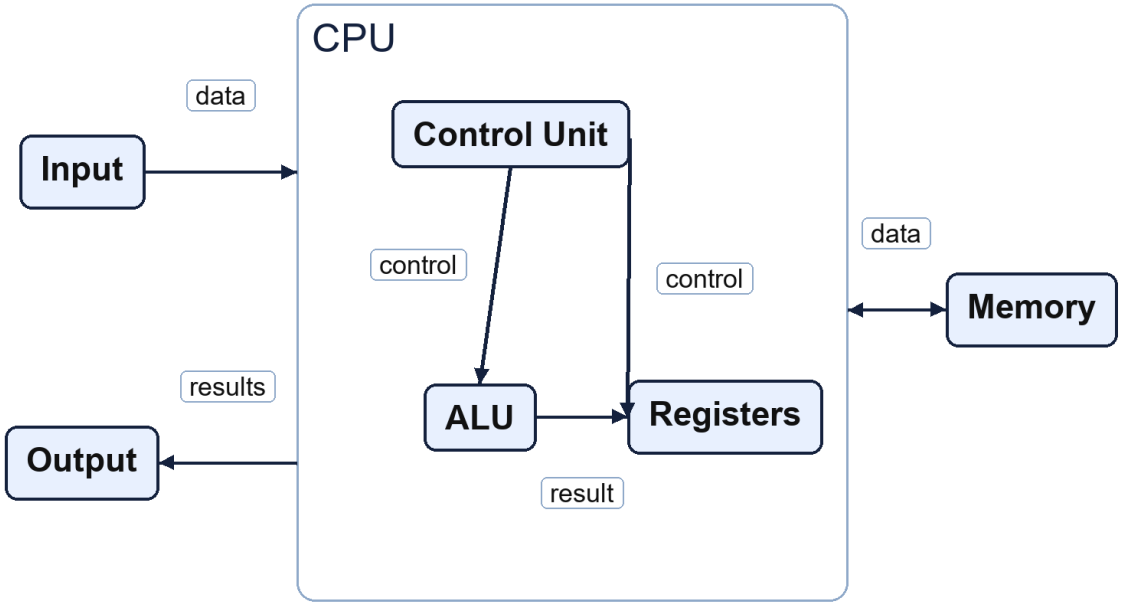



*Figure 1: Stored-program computer organization.*

**(a) (2 marks)** Name the computer organization shown in the figure and state its two key principles.

**Answer:** Von Neumann architecture. Instructions and data are stored in the same memory, and the CPU executes instructions sequentially unless a branch changes the control flow.

**(b) (3 marks)** Explain the function of the control unit and the arithmetic logic unit in this organization.

**Answer:** The control unit fetches and decodes instructions and generates the control signals. The ALU performs arithmetic and logical operations and records status flags.

### QUESTION A2 (5 marks)

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The CPU contains a control unit and an arithmetic logic unit.

**(a) (2 marks)** Explain the function of the control unit.

**Answer:** It fetches and decodes instructions and generates control signals that coordinate the datapath and other components.

**(b) (3 marks)** Explain the function of the ALU.

**Answer:** It performs arithmetic and logical operations on operands and records status flags such as zero, carry, overflow and sign.

**QUESTION A3** **(5 marks)**

Answer the following data-representation questions. Show your complete workflow.

**(a) (2 marks)** Convert 143 to binary. Show your workflow.

**Answer:** 143 = $10001111_2$.

**(b) (3 marks)** Convert $10111101_2$ to decimal. Show your workflow.

**Answer:** $10111101_2 = 189_{10}$.

**QUESTION A4** **(5 marks)**

Answer the following data-representation questions. Show your complete workflow.

**(a) (2 marks)** Represent -21 in 8-bit sign-magnitude form.

**Answer:** 10010101 (sign bit 1, magnitude 21).

**(b) (3 marks)** Represent -21 in 8-bit two's complement form.

**Answer:** 11101011.

**QUESTION A5** **(10 marks)**

Answer the following data-representation questions. Show your complete workflow.

**(a) (2 marks)** Convert 20 to binary.

**Answer:** 20 = $00010100_2$.

**(b) (2 marks)** Convert $00011110_2$ to hexadecimal.

**Answer:** $00011110_2 = 1E_{16}$.

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**(c) (3 marks)** Convert $DE_{16}$ to octal.

**Answer:** $DE_{16} = 336_8$.

**(d) (3 marks)** Convert $276_8$ to decimal.

**Answer:** $276_8 = 190_{10}$.

**QUESTION A6** **(10 marks)**

Study Figure 2 and answer the following questions.

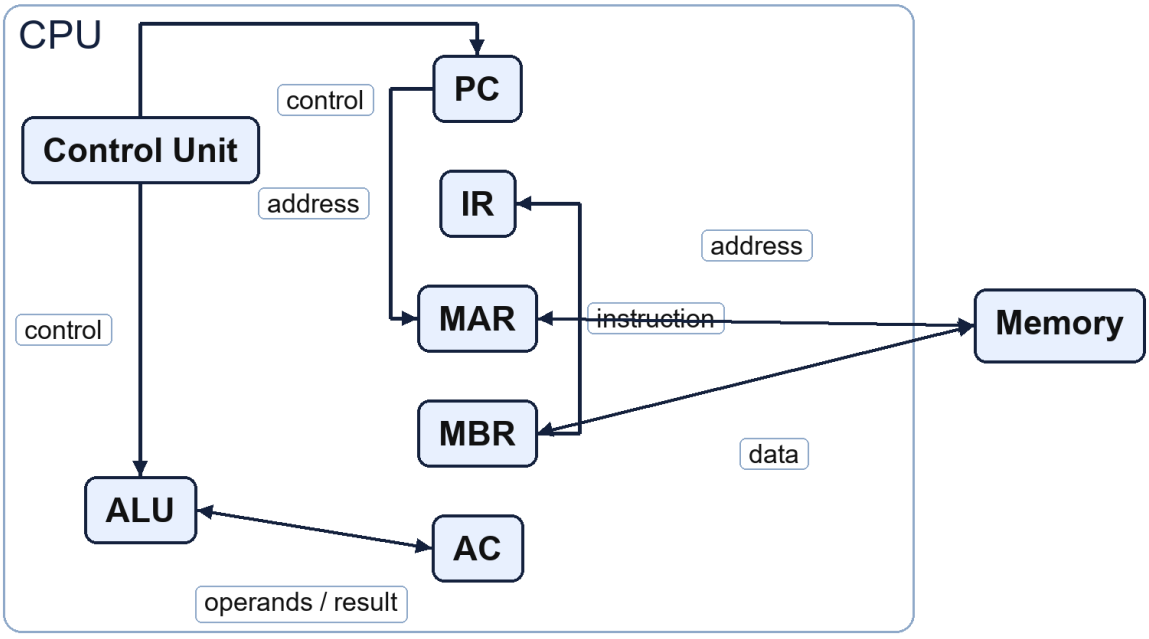



*Figure 2: CPU, memory and system bus organization.*

**(a) (2 marks)** Identify the register that holds the address of the next instruction and the register that holds the instruction currently being decoded.

**Answer:** PC holds the address of the next instruction; IR holds the instruction currently being decoded.

**(b) (3 marks)** Explain the roles of MAR and MBR during a memory read.

**Answer:** The MAR holds the address to be accessed. The memory places the selected word in the MBR, which then supplies it to the CPU (for example, to the IR).

**(c) (5 marks)** Describe the fetch-decode-execute cycle using the components shown in the

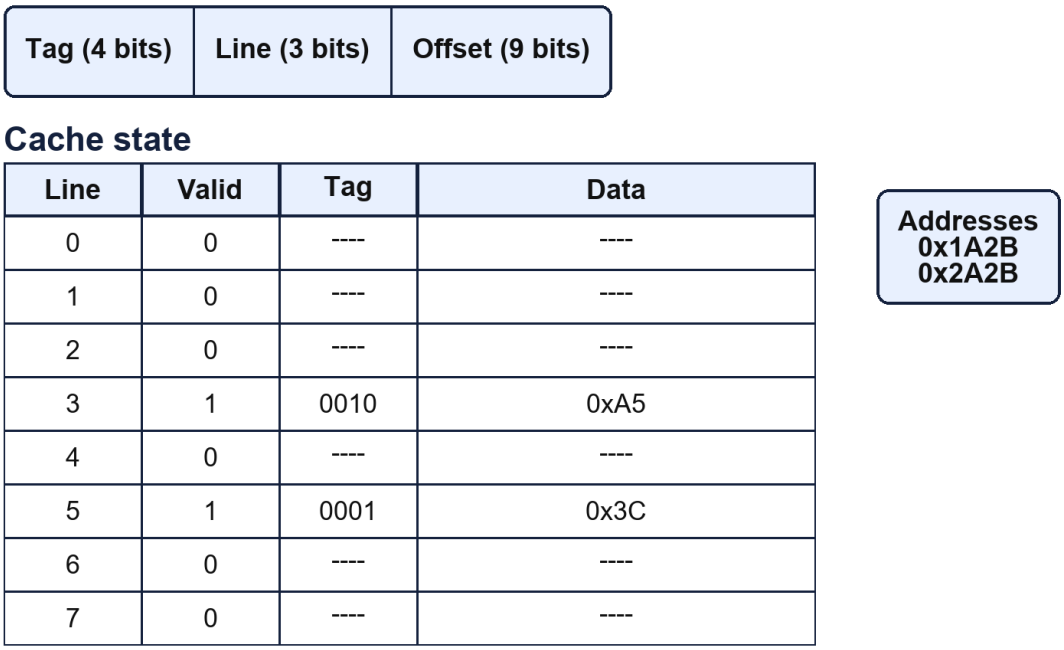
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figure.

**Answer:** Fetch: PC -> MAR; memory read; MBR -> IR; PC is incremented. Decode: the CU interprets the opcode in IR and issues control signals. Execute: operands are read, the ALU performs the operation, the result and flags are written back, and the next instruction is fetched.

**QUESTION A7** **(10 marks)**

Study Figure 3 and answer the following questions.




| Line | Valid | Tag | Data |
|---|---|---|---|
| 0 | 0 | ---- | ---- |
| 1 | 0 | ---- | ---- |
| 2 | 0 | ---- | ---- |
| 3 | 1 | 0010 | 0xA5 |
| 4 | 0 | ---- | ---- |
| 5 | 1 | 0001 | 0x3C |
| 6 | 0 | ---- | ---- |
| 7 | 0 | ---- | ---- |

*Figure 3: Direct-mapped cache organization and address fields.*

**(a) (2 marks)** The cache uses a 16-bit address split into a 4-bit tag, a 3-bit line number and a 9-bit offset. For address 0x1A2B, give the tag, line number and offset in binary.

**Answer:** 0x1A2B = 0001 1010 0010 1011; tag = 0001, line = 101, offset = 000101011.

**(b) (4 marks)** Using the cache state shown in the figure, determine whether each address is a hit or a miss: (i) 0x1A2B, (ii) 0x2A2B. Explain your reasoning.

**Answer:** (i) Hit: tag 0001 and line 101; line 5 is valid with tag 0001. (ii) Miss: tag 0010 and line 101; line 5 is valid but its tag is 0001, so the tags do not match.

**(c) (4 marks)** Explain why the cache compares tags and how temporal and spatial locality

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improve the hit rate.

**Answer:** The tag identifies which memory block occupies a line; comparing tags prevents returning data from a different block. Temporal locality means recently used data is likely reused, and spatial locality means nearby addresses are likely used, so keeping blocks in cache raises the hit rate.

**QUESTION A8** **(10 marks)**

Multiple interrupts can arrive while the CPU is servicing another interrupt.

**(a) (3 marks)** Explain nested interrupts.

**Answer:** A higher-priority interrupt can interrupt a lower-priority ISR; the CPU saves the lower ISR context and services the higher-priority one first.

**(b) (3 marks)** Describe how interrupt priority determines the order of service.

**Answer:** The interrupt controller compares priorities and presents the highest-priority pending request to the CPU first.

**(c) (4 marks)** Explain the sequence of events when a higher-priority interrupt arrives during an ISR.

**Answer:** The CPU finishes the current instruction, saves the current context, acknowledges the higher-priority interrupt and jumps to its ISR; the lower ISR resumes after the higher one returns.

**QUESTION A9** **(10 marks)**

Interrupts can be maskable or non-maskable.

**(a) (3 marks)** Differentiate maskable and non-maskable interrupts.

**Answer:** Maskable interrupts can be disabled/delayed by the CPU; non-maskable interrupts cannot be ignored and are used for critical events.

**(b) (3 marks)** Explain what an interrupt handler (ISR) does.

**Answer:** It saves context, identifies the device/cause, services the request, clears the interrupt and restores context.

**(c) (4 marks)** Describe TWO methods for handling multiple interrupts.

**Answer:** Priority-based nesting and polling/sequential service; nested interrupts allow a higher-priority request to pre-empt a lower-priority ISR.

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**QUESTION A10** **(10 marks)**

Flynn's taxonomy classifies computer architectures by instruction and data streams.

**(a) (4 marks)** Describe SISD, SIMD, MISD and MIMD.

**Answer:** SISD: one instruction/one data stream. SIMD: one instruction/ multiple data streams. MISD: multiple instructions/one data stream. MIMD: multiple instructions/multiple data streams.

**(b) (2 marks)** Give ONE example of SIMD and ONE example of MIMD.

**Answer:** SIMD: GPU/vector processor. MIMD: multicore CPU or cluster.

**(c) (4 marks)** Explain the difference between shared-memory and distributed-memory MIMD.

**Answer:** Shared-memory MIMD uses one address space accessible by all processors; distributed-memory MIMD gives each processor local memory and communicates by messages.

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## PART B — MARKING SCHEME

Worked answers and marking points.

**QUESTION B1** **(20 marks)**

Write appropriate programs using zero-, one- and three-address instructions to compute Z = (A + B) * (C - D) / E. Registers A, B, C, D and E must not be modified; X, Y and Z are temporary registers; the final answer must be stored in Z. Available instructions: zero-address PUSH, POP, ADD, SUB, MUL, DIV; one-address LOAD, STORE, ADD, SUB, MUL, DIV; three-address ADD, SUB, MUL, DIV.

**(a) (6 marks)** Write a zero-address program for the expression.

**Answer:**

```
PUSH A
PUSH B
ADD
PUSH C
PUSH D
SUB
MUL
PUSH E
DIV
POP Z
```

**(b) (6 marks)** Write a one-address program for the expression.

**Answer:**

```
LOAD A
ADD B
STORE X
LOAD C
SUB D
STORE Y
LOAD X
MUL Y
STORE X
LOAD X
DIV E
STORE Z
```

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**(c) (8 marks)** Write a three-address program for the expression.

**Answer:**

```
ADD X, A, B
SUB Y, C, D
MUL X, X, Y
DIV Z, X, E
```

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